

Purplle Online Beauty Shopping Management System

CHAPTER - I

1.0 Introduction

The Purplle Online Beauty Shopping Management System is a database-driven e-commerce application designed to manage the online sale of beauty and personal care products. Purplle is a popular online beauty platform that offers a wide range of skincare, makeup, haircare, fragrances, and wellness products from multiple brands.

This project aims to develop an efficient database management system that stores and manages customer information, product details, inventory, orders, payments, and delivery records. It enables customers to browse products, place orders, make secure payments, and track their purchases, while administrators can efficiently manage products, stock, customer information, and sales reports.

The project demonstrates how a relational database supports an e-commerce platform by ensuring data accuracy, reducing redundancy, maintaining data integrity, and improving business operations.

1.1 Objective

The primary objective of this project is to design and implement a database management system for the Purplle Online Beauty Shopping platform that efficiently manages customers, products, orders, inventory, payments, and deliveries while providing a secure and user-friendly shopping experience.

1.2 Key Objectives

- To design a normalized relational database for an online beauty shopping platform.
- To maintain customer registration and login information.
- To manage beauty product details and categories.
- To maintain accurate inventory and stock availability.
- To process customer orders efficiently.
- To record payment transactions securely.
- To manage shipping and delivery details.
- To generate sales and inventory reports.

- To reduce data redundancy and improve data consistency.
- To ensure secure storage of customer and business data.

1.3 Motive of the Project

The motive of this project is to simplify and automate the management of an online beauty shopping platform through an efficient database management system. Manual handling of customer records, product inventories, and orders is time-consuming and prone to errors.

This project helps improve operational efficiency by organizing data systematically, reducing duplicate records, ensuring faster order processing, maintaining accurate inventory, and providing customers with a smooth online shopping experience.

1.4 Problem Statement

Online beauty shopping platforms handle thousands of customers, products, and transactions every day. Managing this information manually can lead to errors, duplicate records, stock mismatches, delayed order processing, and poor customer service.

The Purple Online Beauty Shopping Management System addresses these issues by providing a centralized database that manages customer information, products, inventory, orders, payments, and deliveries efficiently. The system ensures accurate data management, faster processing, improved security, and better customer satisfaction.

1.5 Business Requirements

The system should support the following business requirements:

- Maintain customer registration and profile information.
- Store complete product details with categories and prices.
- Track inventory and stock availability.
- Allow customers to search and purchase products online.
- Process customer orders efficiently.
- Support secure payment processing.
- Generate invoices after successful purchases.
- Track shipping and delivery status.
- Manage product reviews and ratings.

- Generate sales, customer, and inventory reports.
- Maintain data security and privacy.

1.6 Functional Requirements

- Customer registration and login.
- Browse and search products.
- Add products to cart.
- Place and manage orders.
- Secure online payment.
- Track order status.
- Manage products and inventory.
- Generate sales reports.

1.7 Non-Functional Requirements

- Fast system performance.
- Secure user and payment data.
- High system reliability.
- User-friendly interface.
- 24/7 system availability.
- Easy maintenance and updates.
- Scalable for future growth.
- Accurate and consistent data.

1.8 Stakeholders

- **Customers** – Browse products, place orders, make payments, track deliveries, and provide reviews.
- **Administrator** – Manage products, inventory, customer records, orders, payments, and reports.
- **Suppliers** – Supply beauty products and update product availability.

- **Delivery Partners** – Deliver orders and update delivery status.
- **Payment Gateway** – Process secure online payments and refunds.
- **Company Management** – Monitor sales and make business decisions.
- **Customer Support** – Resolve customer queries, returns, and complaints.

1.9 Conclusion

The **Purple Online Beauty Shopping Management System** is designed to efficiently manage online beauty product sales by organizing customer details, product information, orders, payments, and inventory in a structured database. The system reduces manual work, improves data accuracy, ensures secure transactions, and enhances the overall shopping experience. It also helps administrators manage business operations efficiently and supports better decision-making through organized data and reports. Overall, this project demonstrates how a **Database Management System (DBMS)** can improve the efficiency and reliability of an e-commerce platform.